Data Wrangling 🤠

Lists

Lesson 9

Many Data Points

- Tedious to create a variable for every data point we want to operate on

```python
# conversions.py
temp_1 = 0.0 # in Celsius
temp_2 = 20.0
temp_3 = 37.0
```

Lists

- A **list** is an ordered sequence of values
- Each value is called an **element**, addressed by its **index**
- We can add and remove values from a list at runtime

```
temps = [0.0, 20.0, 37.0, 15.5, 50.0]
```

Creating a List

- **Empty list**: add elements later

```
temps = []
```

- **List literal**: list with values specified upon intialization

```
temps = [0.0, 20.0, 37.0, 15.5, 50.0]
names = ["Vrinda", "Alicia", "Taylor"]
```

Demo: Accessing Elements

- Use square brackets to read a value – indices start at `0`

```
>>> names = ["Vrinda", "Alicia", "Taylor"]
>>> names[0]
'Vrinda'
>>> names[1]
'Alicia'
```

- How do we determine the last index?

```
>>> names[len(names) - 1]
'Taylor'
```

Demo: Updating Elements

- Assign directly to an index to change a value

```
>>> names = ["Vrinda", "Alicia", "Taylor"]
>>> names[1] = "Tevin"
>>> names
["Vrinda", "Tevin", "Taylor"]
```

Demo: Appending Elements

- `append` adds an element to the **end** of the list

```
>>> names.append("Mari")
>>> names
["Vrinda", "Tevin", "Taylor", "Mari"]
>>> names.append("Alicia")
>>> names
["Vrinda", "Tevin", "Taylor", "Mari", "Alicia"]
>>> pets = []
>>> pets.append("Daisy")
>>> pets.append("Ollie")
>>> pets
["Daisy", "Ollie"]
```

Demo: Removing Elements

- `pop` removes and returns the element at a given index

```
>>> names.pop(1)
>>> names
["Vrinda", "Taylor", "Mari", "Alicia"]
>>> names.pop()
>>> names
["Vrinda", "Taylor", "Mari"]
```

List Operations

Operation	Form	Result
Length	`len(list_name)`	returns list length
Access	`list_name[index]`	access at index
Update	`list_name[index] = value`	updates value at index
Append	`list_name.append(value)`	adds value to end
Pop	`list_name.pop(index)`	removes at index

Demo: Lists + While Loops

```python
# lists.py
temps = [0.0, 20.0, 37.0]
converted_temps = []

i = 0

while i < len(temps):
    temp = temps[i]
    temp_in_f = celsius_to_fahrenheit(temp)
    converted_temps.append(temp_in_f)
    print(converted_temps[i])
    i = i + 1
```